

Building Deep Res

Ben Batorsky + Eric Ma

repo: <https://github.com/ericmjl/build-deep-research-agent>

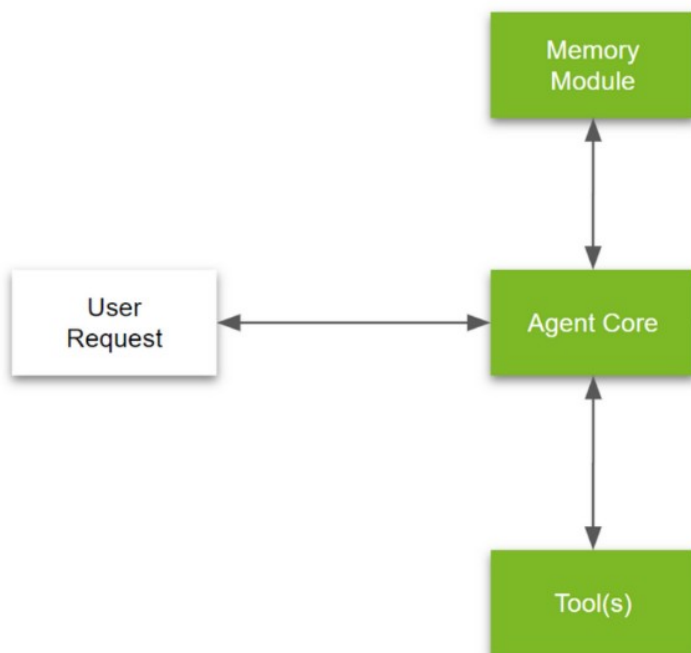
Follow setup instructions, run `pixi run nb0` for initial setup



- PhD, Policy Analysis/Economics, RAND Corporation
- Decade in Data Science, focus on NLP
- US Responsible AI Lead, Philips

- ScD, B
- <https://>
- Traine
- Senior





What should we focus on?

What do you want to research?



Deep research

Pro



Sources

Files



Sugge

↗ Ev

↗ Tra

↗ Co

↗ Ev



Ask for research on "pitch"



Researching websites...



mitco...

The Heilmeier Ca...



csee.u...

DARPA Heilmeyer...



en.wiki...

George H. Heilm...



umass...

3MT Competition ...



three...

Competition rule...



ucm.es

A simple templat...





Example basic prompt

(Ollama->Llama3.2)

<|start_header_id|>system<|end_header_id|> You are a helpful assistant.

<|start_header_id|>user<|end_header_id|> Hi!<|eot_id|>

<|start_header_id|>assistant<|end_header_id|>

Exam

<|start_header_id|>

When you
answer to

You are a

<|eot_id|>

Given the
function of
prompt.

Identity
Gives the model its
identity and constraints

Instructions
Directs the model on what
to do

Examples
Patterns showing how the
model should respond

Context
Relevant information the
model should use









- Security - Could the agent be misused?
- Reliability - Can I trust what the agent produces?
- Ethics - Is the output intellectually honest?









Risk

Mitigation

Security

Least privilege, sandboxing, verification

Reliability

pass $\wedge$ k, groundedness

Ethics

System cards, community involvement